

Joseph Zhang

Senior Data Engineer / ML Engineer

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Profile

Results-driven **Senior Data Engineer** and **ML Engineer** with **10 years** of experience designing, building, and optimizing scalable data and machine learning systems. Skilled in developing automated ETL pipelines, managing cloud-based data infrastructures (**Azure**, **AWS**), and deploying machine learning models at scale. Strong background in data architecture, **Airflow** orchestration, and distributed data engineering systems using **Spark**, **Kafka**, **Hadoop**, and **dbt**. Known for delivering efficient, data-driven solutions that accelerate business growth, reduce operational overhead, and enhance decision-making. Adept at mentoring teams and collaborating across technical and business domains to deliver measurable impact.

Professional Experience

Senior Data Engineer, Accenture

10/2023 – Present | Chicago, IL

- Led the data transformation process with **DBT** and **Snowflake**, ensuring high-quality, structured data for analytics, reporting, and machine learning model development.
- Designed and built a cloud-based data pipeline integrating **Salesforce** and **SAP**, enabling seamless on-prem to cloud migration using **Azure Data Factory** and **CDC**, improving data integration efficiency by 35%.
- Built and optimized ETL workflows to transform raw data into **ODS**, **DWH**, and **Data Marts**, improving enterprise analytics and reporting processes across the organization.
- Improved data query performance by implementing **Azure Synapse Analytics** with serverless SQL, enabling high-speed analytics on structured and semi-structured data in **Azure Data Lake Storage** and reducing processing time by 30%.
- Built data ingestion and analytics workflows using **Microsoft Fabric Lakehouse** and Data Pipelines, enabling centralized storage in OneLake and simplifying downstream reporting and analytics.
- Developed high-performance, scalable data pipelines using **Apache Spark** and **Azure Databricks**, achieving a 40% reduction in processing time for both batch and real-time data, ensuring faster insights delivery.
- Implemented real-time data streaming solutions with **Azure Event Hubs** and **Azure Stream Analytics**, ensuring immediate data validation and integrity checks to improve the reliability and operational efficiency of data pipelines.
- Collaborated closely with engineers, analysts, and business teams to ensure smooth deployment of data pipelines through **CI/CD** processes, enhancing overall workflow and delivery speed.
- Mentored junior team members, providing guidance on best practices for **ETL** automation, cloud technologies, and scalable data pipeline architectures.

Data Engineer/ML Engineer, Walmart

03/2020 – 09/2023 | Bentonville, AR

- Architected and optimized data pipelines using **Azure Data Factory** with **Azure Synapse Pipelines**, improving performance by 40% and enabling scalable, enterprise-grade data workflows.

- Implemented data transformation and modeling frameworks using **dbt**, enabling modular, version-controlled **SQL** workflows and improving data reliability and documentation across analytics environments.
- Integrated sales forecasting with business intelligence by building **Power BI** dashboards and **Tableau** reports, providing actionable insights for the sales and supply chain teams, which improved forecast accuracy by **25%**.
- Built and deployed predictive models with **LLMs** to enhance data-driven decision-making, ensuring sensitive data is classified and protected according to legal and regulatory standards, reducing data breaches.
- Developed and deployed a predictive model using **LSTM**, **GANs**, and **TensorFlow** to automate behavior analysis, reducing the operational workload and increasing team efficiency.
- Containerized machine learning models using **Kubernetes** and **Docker**, ensuring efficient deployment and scalability of ML models across cloud environments, leading to a **40%** improvement in deployment time and model scalability.
- Developed and deployed machine learning models using **Amazon SageMaker**, streamlining predictive analytics for customer behavior analysis, which resulted in a **20%** improvement in marketing campaign effectiveness and sales forecasting accuracy.

Data Engineer, Global IT Associates

01/2016 – 02/2020 | Jacksonville, FL

- Designed and implemented automated test suites using **Python** and **SQL** to validate the accuracy and consistency of data being migrated from the **Guidewire** database to the **DWH**.
- Leveraged **AWS Glue** for orchestration, ensuring high-quality, reliable, and error-free data migration pipelines.
- Built and managed scalable data warehouses using **Amazon S3** and **Amazon Redshift**, achieving a 40% reduction in query latency and significantly improving data retrieval speed for reporting and analytics.
- Optimized the transformation of large-scale datasets from the **DWH** into **Data Marts** using **Amazon EMR**, enhancing data processing efficiency and reducing transformation time by 20%, enabling faster insights and decision-making.
- Built and automated complex **ETL** workflows using **Apache Airflow** to orchestrate data pipelines, integrating data from various insurance systems and boosting data processing efficiency by **30%**, improving the timeliness of business-critical data availability.
- Designed and optimized scalable data pipelines using **Apache Kafka**, enabling efficient **ETL** processes and real-time data processing, which enhanced analytics and reporting capabilities, ensuring that stakeholders had access to up-to-date, actionable insights.

Software Engineer, sibedge

09/2014 – 12/2015 | Cupertino, CA

- Developed and optimized full-stack web applications using **Python** for backend development and **HTML**, **CSS**, and **JavaScript** for frontend development, creating seamless and responsive user experiences across both web and mobile platforms.
- Designed and maintained **RESTful APIs** to interact with a cloud-based backend, ensuring scalability, security, and data consistency, while handling large volumes of requests from multiple clients.
- Implemented backend systems using **MySQL**, **PostgreSQL**, and **Oracle** for relational data management, while optimizing performance with **MongoDB** and **MariaDB** to ensure efficient data storage and retrieval, significantly improving database query speeds.

- Conducted regular database maintenance and optimization tasks using **SSMS**, including indexing, query optimization, and analyzing execution plans, which resulted in improved database performance and reduced load times.

Education

Bachelor's Degree in Computer Science,
Illinois Institute of Technology

01/2010 – 05/2014 | Chicago, IL

Skills

Programming & Development

Python, SQL, R, Bash, PySpark, Scikit-Learn, TensorFlow, PyTorch, Pandas, NumPy, Version Control, (Git, GitHub, Bitbucket), CI/CD Pipelines, Agile / Scrum

Cloud Platforms

- **Azure:** Blob Storage, Data Factory, Databricks, Synapse Analytics, Data Lake
- **AWS:** S3, Redshift, Glue, EMR, Kinesis, Lambda, SageMaker, EC2, RDS, IAM

Data Engineering & Big Data

- **ETL / Orchestration:** Apache Airflow, Azure Data Factory, AWS Glue
- **Big Data Processing:** Apache Spark, Azure Databricks, Amazon EMR
- **Streaming & Messaging:** Apache Kafka, Azure Event Hubs, Amazon Kinesis
- **Data Warehousing:** Azure Synapse Analytics, Snowflake, Azure SQL Database, Amazon Redshift
- **Data Modeling:** Kimball, Inmon, Data Vault methodologies
- **Databases:** MySQL, PostgreSQL, Redis, MongoDB, Oracle, SQL Server

Machine Learning & AI

Predictive Modeling, Deep Learning, LLMs, NLP, Computer Vision, Feature Engineering, Model Deployment (Kubernetes, SageMaker), Prompt Engineering

DevOps & Infrastructure

- **Containerization:** Docker, Kubernetes
- **Infrastructure as Code:** Terraform, CloudFormation
- **Monitoring & CI/CD:** Jenkins, GitHub Actions
- **Security & Governance:** IAM, Data Encryption, Compliance Automation

Visualization & Collaboration

- Power BI, Tableau
- Jira, Confluence, Slack, Scrum, Sprint Planning